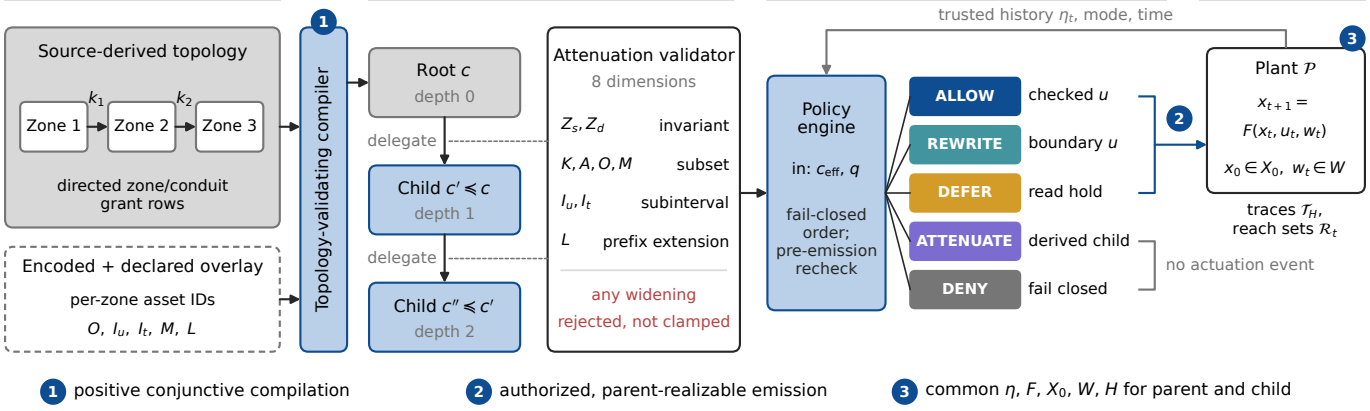
Policy input
Capability lineage
Runtime authorization
Plant
Source-derived topology
Zone 1
Zone 2
Zone 3
$k_1$
$k_2$
directed zone/conduit grant rows
Encoded + declared overlay
per-zone asset IDs
$O$, $I_u$, $I_t$, $M$, $L$
1
Topology-validating compiler
Root $c$
depth 0
delegate
Child $c' \preccurlyeq c$
depth 1
delegate
Child $c'' \preccurlyeq c'$
depth 2
Attenuation validator
8 dimensions
$Z_s, Z_d$ invariant
$K, A, O, M$ subset
$I_u, I_t$ subinterval
$L$ prefix extension
any widening rejected, not clamped
trusted history $\eta_t$, mode, time
Policy engine
in: $c_{\text{eff}}$, $q$
fail-closed order; pre-emission recheck
ALLOW
checked $u$
REWRITE
boundary $u$
DEFER
read hold
ATTENUATE
derived child
DENY
fail closed
2
no actuation event
3
Plant $\mathcal{P}$
$x_{t+1} =$
$F(x_t, u_t, w_t)$
$x_0 \in X_0,\ w_t \in W$
traces $\mathcal{T}_H$,
reach sets $\mathcal{R}_t$
1 positive conjunctive compilation
2 authorized, parent-realizable emission
3 common $\eta$, $F$, $X_0$, $W$, $H$ for parent and child